Solution Requirements Specification

Functional and non-functional requirements

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 4

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Functional Requirements

ID	Category	Requirement	Priority
FR-01	Registration	Register unique user; hash password with bcrypt before persistence.	High
FR-02	Authentication	Validate login and issue signed 30-day JWT.	High
FR-03	Profile	Return authenticated profile only when valid Bearer token is supplied.	High
FR-04	Weather	Retrieve city weather through OpenWeatherMap with deterministic fallback.	High
FR-05	Locations	Create/list/update/delete favorite locations with user ownership.	High
FR-06	AI Summary	Convert weather metrics into concise natural-language explanation.	High
FR-07	AI Recommendation	Generate outfit, commute and hydration recommendations.	High
FR-08	Validation	Return predictable status codes and JSON error structures.	Medium

Non-Functional Requirements

ID	Quality	Requirement / Target
NFR-01	Security	Salted bcrypt password hashing; signed JWT; protected route middleware.
NFR-02	Reliability	External weather and AI failures must have local fallback paths.
NFR-03	Performance	Target responsive API behavior; benchmark representative endpoints and AI latency.
NFR-04	Scalability	Services should be separable and cacheable; future Redis/cloud scaling is planned.
NFR-05	Maintainability	MVC separation, reusable services, centralized error handling and environment configuration.
NFR-06	Data integrity	Unique email and compound unique location index prevent inconsistent records.

Acceptance Criteria

- Every endpoint has a defined method, access level and expected purpose.
- Protected location operations cannot cross user boundaries.
- A weather result is still available when OpenWeatherMap is unavailable.
- An AI summary/recommendation can degrade to deterministic local rules.
- The complete test collection can be executed by a new evaluator using documented environment variables.